ISE LABS

HTOL Setup Information

For

Marvell

DEVICE #: Iliad

DEVICE REV: A
PACKAGE: 1989 HFCBGA 0.8mm pitch
HTOL Equipment: Sonoma

Last Updated: 08/27/2025
Prepared by: Aaron Graf

ISE LABS	HTOL setup information for Marvell Iliad

B/I Pattern information:

Pattern Name	Stress Time (minutes)
all_fastio_burnin_01_23DG.hex	5
iliada0_ilcq_mbist_mbist_burnin_stagger_ffff_r500s500t500d500cc500cd500_v0_20241218DG.hex	5
iliada0_ilcq_mbist_pbist_memtype1_r2550s1050t3150d1600q2000cc1800cd2400_v0_20250416DG.hex	0
iliada0_ilall_burnin_scan_burnin_v0_20240829DG.hex	5

Device and Board Information

- 6 DUT Positions per board

HTOL Chamber Temperature:
Ta = N/A
Tc = 105C

Power Supply Information:

Power Domain	Supply #	Voltage	Seq Up	Seq Down	V Set Limits
VDDC	CORE1/3/5	1.044V	1	8	0.8V – 1.19V
VDD_DIG	VOUT17-22	0.75V	2	7	0.55V – 0.95V
VDD_IO	VOUT14	1.98V	3	6	1.5V – 2.3V
VDD_HA	VOUT1-6	1.32V	4	5	1.1V – 1.5V
VDD_DDR1	VOUT9/11	1.21V	5	4	1.0V – 1.4V
VDD_DDR0	VOUT10/12	1.21V	6	3	1.0V – 1.4V
VDD_DIG_QIP01	CORE2/4/6	0.825V	7	2	0.39V – 0.99V
VDD_DIG_DDR01	VOUT13/14/16	0.85V	8	1	0.6V – 1.0V

Currents

Power Domain	Supply #	Voltage	DUT CURRENT
VDDC	CORE1/3/5	1.044V	103A
VDD_DIG	VOUT17-22	0.75V	3.9A
VDD_IO	VOUT14	1.98V	0.1A
VDD_HA	VOUT1-6	1.32V	3.2A
VDD_DDR1	VOUT9/11	1.21V	1.5A
VDD_DDR0	VOUT10/12	1.21V	1.8A
VDD_DIG_QIP01	CORE2/4/6	0.825V	25A
VDD_DIG_DDR01	VOUT13/14/16	0.85V	11A

Expected Currents

ISE LABS

HTOL
setup information for
Marvell Iliad

Power Domain	Supply #	Voltage	CURRENT
VDDC	CORE1/3/5	1.044V	117A
VDD_DIG	VOUT17-22	0.75V	9A
VDD_IO	VOUT14	1.98V	2A
VDD_HA	VOUT1-6	1.32V	6A
VDD_DDR1	VOUT9/11	1.21V	11A
VDD_DDR0	VOUT10/12	1.21V	11A
VDD_DIG_QIP01	CORE2/4/6	0.825V	40A
VDD_DIG_DDR01	VOUT13/14/16	0.85V	11A

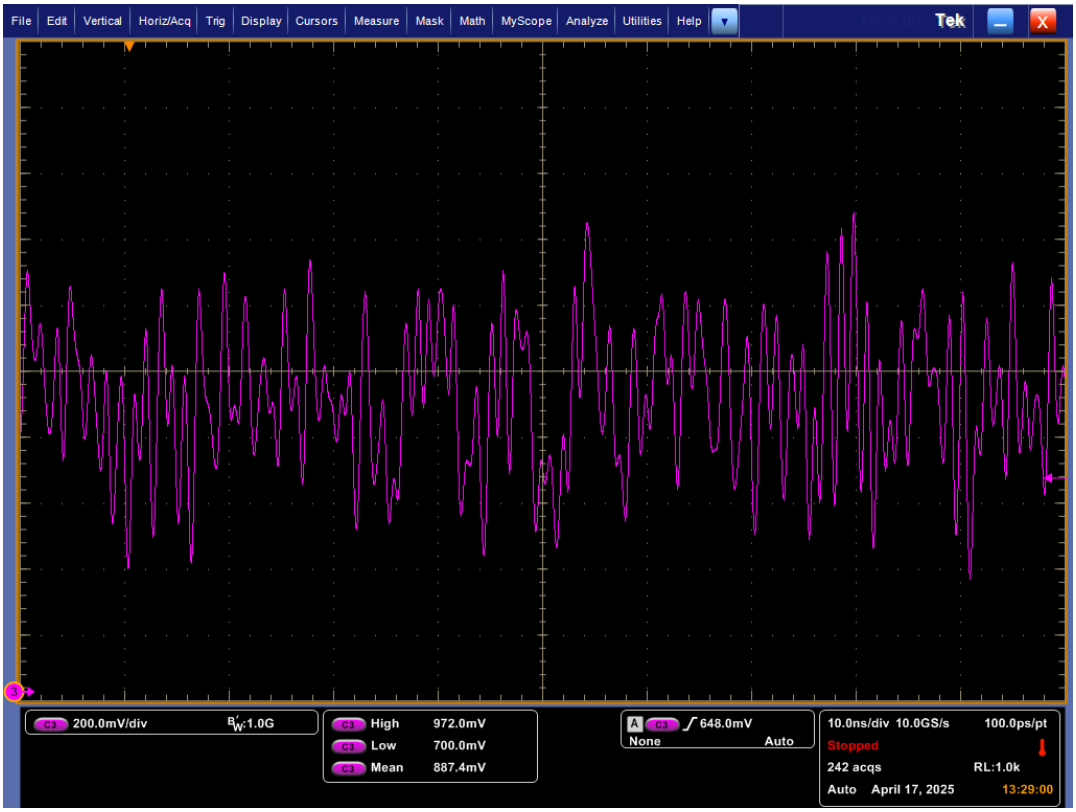
Current Limits Set

Power Domain	Supply #	Voltage	Current Limit set	Duts per Supply
VDDC	CORE1/3/5	1.044V	120A	1
VDD_DIG	VOUT17-22	0.75V	12A	1
VDD_IO	VOUT14	1.98V	40A	6
VDD_HA	VOUT1-6	1.32V	12A	1
VDD_DDR1	VOUT9/11	1.21V	40A	3
VDD_DDR0	VOUT10/12	1.21V	40A	3
VDD_DIG_QIP01	CORE2/4/6	0.825V	100A	1
VDD_DIG_DDR01	VOUT13/14/16	0.85V	45A	2

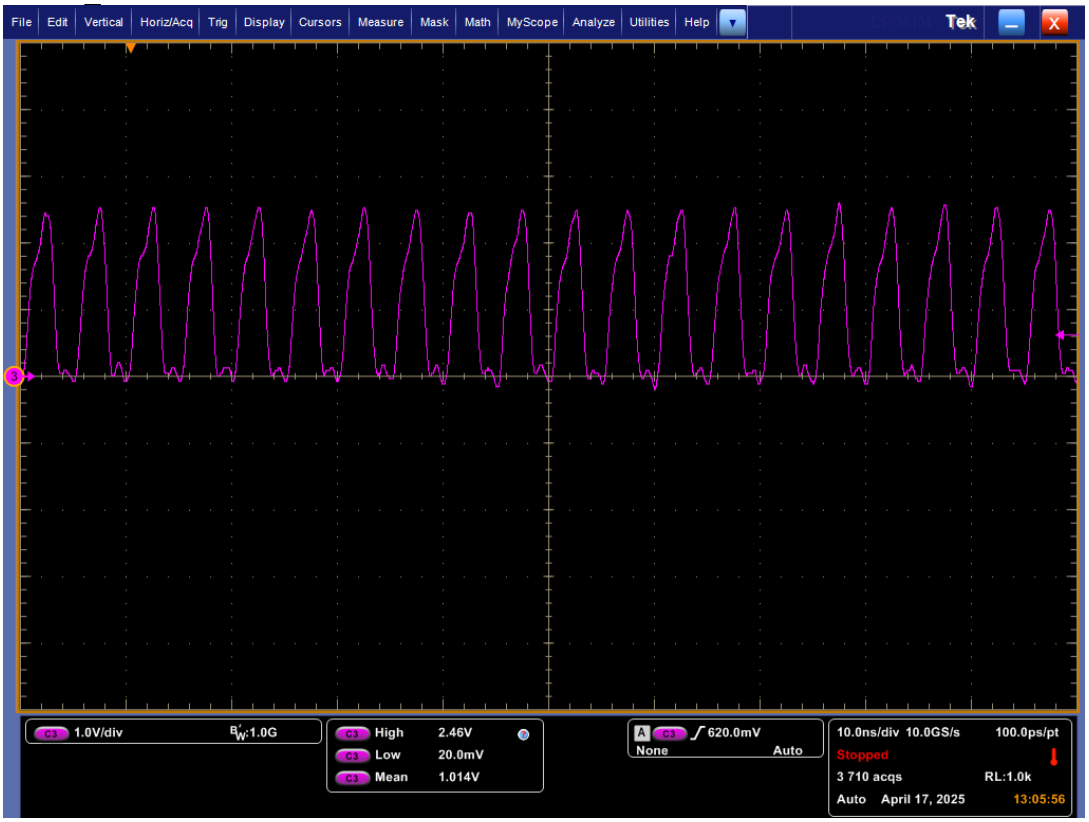
Voltage References

Bank	High level
1	2.0 V
2	2.0 V
3	2.0 V
4	2.0 V

Expected Outputs FASTIO DDR0/1/2/3



GPIO_10



ISE LABS

HTOL
setup information for
Marvell Iliad

GSERP

OUTPUT LOG

```
DEBUG(vectors/iliada0_ilall_burnin_all_fastio_burnin_v1_20250404DG.hex): vector_count = 5047384
Vector Run Time=0.000000 DEBUG:1: start=        0 vec_count=5047385
VECTOR FAILED: error_count=56
Vector# = 2913767 Cycle# = 04082405 Pin(s) Failed = JTG_TDO_C1
Vector# = 2913774 Cycle# = 04082412 Pin(s) Failed = JTG_TDO_C1
Vector# = 2914496 Cycle# = 04086128 Pin(s) Failed = JTG_TDO_C1
Vector# = 2914503 Cycle# = 04086135 Pin(s) Failed = JTG_TDO_C1
Vector# = 2915225 Cycle# = 04089851 Pin(s) Failed = JTG_TDO_C1
Vector# = 2915232 Cycle# = 04089858 Pin(s) Failed = JTG_TDO_C1
Vector# = 2915954 Cycle# = 04093574 Pin(s) Failed = JTG_TDO_C1
Vector# = 2915961 Cycle# = 04093581 Pin(s) Failed = JTG_TDO_C1
Vector# = 5022600 Cycle# = 51340706 Pin(s) Failed = JTG_TDO_C1
Vector# = 5022780 Cycle# = 51340886 Pin(s) Failed = JTG_TDO_C1
Vector# = 5022960 Cycle# = 51341066 Pin(s) Failed = JTG_TDO_C1
Vector# = 5023140 Cycle# = 51341246 Pin(s) Failed = JTG_TDO_C1
Vector# = 5024866 Cycle# = 51342972 Pin(s) Failed = JTG_TDO_C1
Vector# = 5025046 Cycle# = 51343152 Pin(s) Failed = JTG_TDO_C1
Vector# = 5025226 Cycle# = 51343332 Pin(s) Failed = JTG_TDO_C1
Vector# = 5025406 Cycle# = 51343512 Pin(s) Failed = JTG_TDO_C1
Vector# = 5027132 Cycle# = 51345238 Pin(s) Failed = JTG_TDO_C1
Vector# = 5027312 Cycle# = 51345418 Pin(s) Failed = JTG_TDO_C1
Vector# = 5027492 Cycle# = 51345598 Pin(s) Failed = JTG_TDO_C1
Vector# = 5027672 Cycle# = 51345778 Pin(s) Failed = JTG_TDO_C1
```

Vector# = 5029398 Cycle# = 51347504 Pin(s) Failed = JTG_TDO_C1
Vector# = 5029578 Cycle# = 51347684 Pin(s) Failed = JTG_TDO_C1
Vector# = 5029758 Cycle# = 51347864 Pin(s) Failed = JTG_TDO_C1
Vector# = 5029938 Cycle# = 51348044 Pin(s) Failed = JTG_TDO_C1
Vector# = 5035474 Cycle# = 51853579 Pin(s) Failed = JTG_TDO_C1
Vector# = 5035475 Cycle# = 51853580 Pin(s) Failed = JTG_TDO_C1
Vector# = 5035654 Cycle# = 51853759 Pin(s) Failed = JTG_TDO_C1
Vector# = 5035655 Cycle# = 51853760 Pin(s) Failed = JTG_TDO_C1
Vector# = 5035834 Cycle# = 51853939 Pin(s) Failed = JTG_TDO_C1
Vector# = 5035835 Cycle# = 51853940 Pin(s) Failed = JTG_TDO_C1
Vector# = 5036014 Cycle# = 51854119 Pin(s) Failed = JTG_TDO_C1
Vector# = 5036015 Cycle# = 51854120 Pin(s) Failed = JTG_TDO_C1
Vector# = 5036607 Cycle# = 51854712 Pin(s) Failed = JTG_TDO_C1
Vector# = 5036608 Cycle# = 51854713 Pin(s) Failed = JTG_TDO_C1
Vector# = 5036787 Cycle# = 51854892 Pin(s) Failed = JTG_TDO_C1
Vector# = 5036788 Cycle# = 51854893 Pin(s) Failed = JTG_TDO_C1
Vector# = 5036967 Cycle# = 51855072 Pin(s) Failed = JTG_TDO_C1
Vector# = 5036968 Cycle# = 51855073 Pin(s) Failed = JTG_TDO_C1
Vector# = 5037147 Cycle# = 51855252 Pin(s) Failed = JTG_TDO_C1
Vector# = 5037148 Cycle# = 51855253 Pin(s) Failed = JTG_TDO_C1
Vector# = 5037740 Cycle# = 51855845 Pin(s) Failed = JTG_TDO_C1
Vector# = 5037741 Cycle# = 51855846 Pin(s) Failed = JTG_TDO_C1
Vector# = 5037920 Cycle# = 51856025 Pin(s) Failed = JTG_TDO_C1
Vector# = 5037921 Cycle# = 51856026 Pin(s) Failed = JTG_TDO_C1
Vector# = 5038100 Cycle# = 51856205 Pin(s) Failed = JTG_TDO_C1
Vector# = 5038101 Cycle# = 51856206 Pin(s) Failed = JTG_TDO_C1
Vector# = 5038280 Cycle# = 51856385 Pin(s) Failed = JTG_TDO_C1
Vector# = 5038281 Cycle# = 51856386 Pin(s) Failed = JTG_TDO_C1
Vector# = 5038873 Cycle# = 51856978 Pin(s) Failed = JTG_TDO_C1
Vector# = 5038874 Cycle# = 51856979 Pin(s) Failed = JTG_TDO_C1
Vector# = 5039053 Cycle# = 51857158 Pin(s) Failed = JTG_TDO_C1
Vector# = 5039054 Cycle# = 51857159 Pin(s) Failed = JTG_TDO_C1
Vector# = 5039233 Cycle# = 51857338 Pin(s) Failed = JTG_TDO_C1
Vector# = 5039234 Cycle# = 51857339 Pin(s) Failed = JTG_TDO_C1
Vector# = 5039413 Cycle# = 51857518 Pin(s) Failed = JTG_TDO_C1
Vector# = 5039414 Cycle# = 51857519 Pin(s) Failed = JTG_TDO_C1
2025-04-07_13-00-18 Total Time=11.000000 Pattern: 1 Total Loops:1 Average Failures:56 Max Failures:56

Expected Outputs MBIST
TDO has 2 or 3 failing vectors

Expected Outputs SCAN
TDO has one failing vector
DDR pins are scan outputs and toggle